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Introduction

This quick-start guide covers:

- How to start the calibration process
- How to calibrate cameras with the active wand (*IR version 2*)
- Effective wand-waving technique
- Calibration feedback and camera health (*in brief*)
- How to confirm a successful calibration

Important - this guide assumes you have already set your camera masks.
If you have not yet masked your cameras, please complete this through the steps in one of the following guides:

- Automatic masking
- Manual masking

This is the second of the three-part quick-start system preparation guide:

1. Masking your Cameras
2. ***Calibrating your Cameras***

3. Setting Volume Origin and Plane

Camera Calibration (Quick-Start Guide)

In order for you to begin capturing your subject's 3D motion tracking data, the cameras need to be calibrated to the environment you are working in.

The cameras are very sensitive to changes in temperature and light, as well as being physically disturbed. As such it is recommended that you calibrate at the beginning of any new day or capture session, and re-calibrate if the environment has changed or been disturbed.

Calibration is what allows each camera to determine the position of a tracked marker relative to itself, and the position of other cameras relative to itself. This positional data is synchronised between every camera in order to determine the 3D space (*the volume*) and where within that volume that the marker currently is.

Calibration Setup

System State and Panel View

In order to calibrate your cameras you must first ensure that you are in the correct system state.

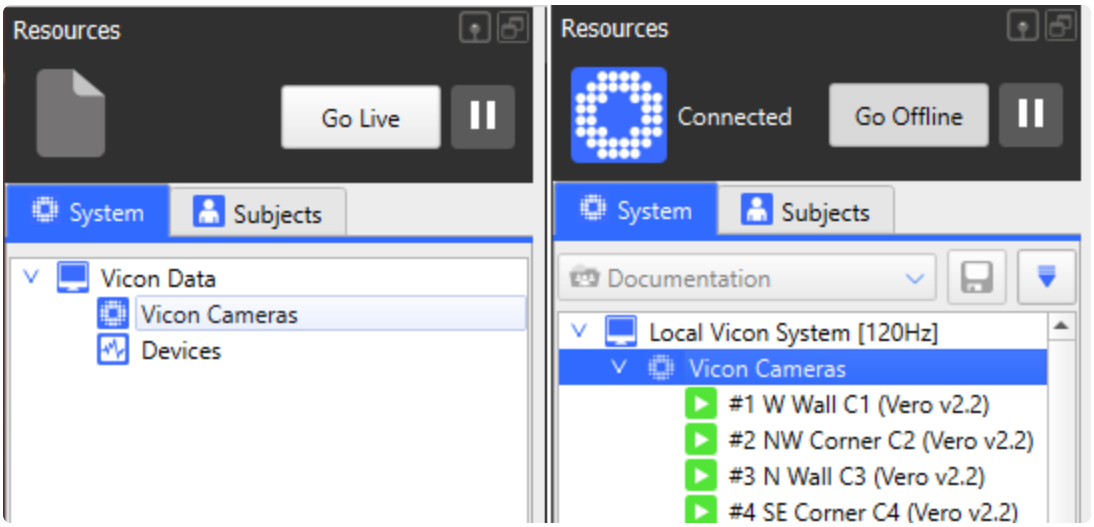
Offline mode supports:

1. subject preparation
2. labelling and editing
3. trial pipelines

Live mode supports:

1. system preparation
2. subject capture
3. trial capture

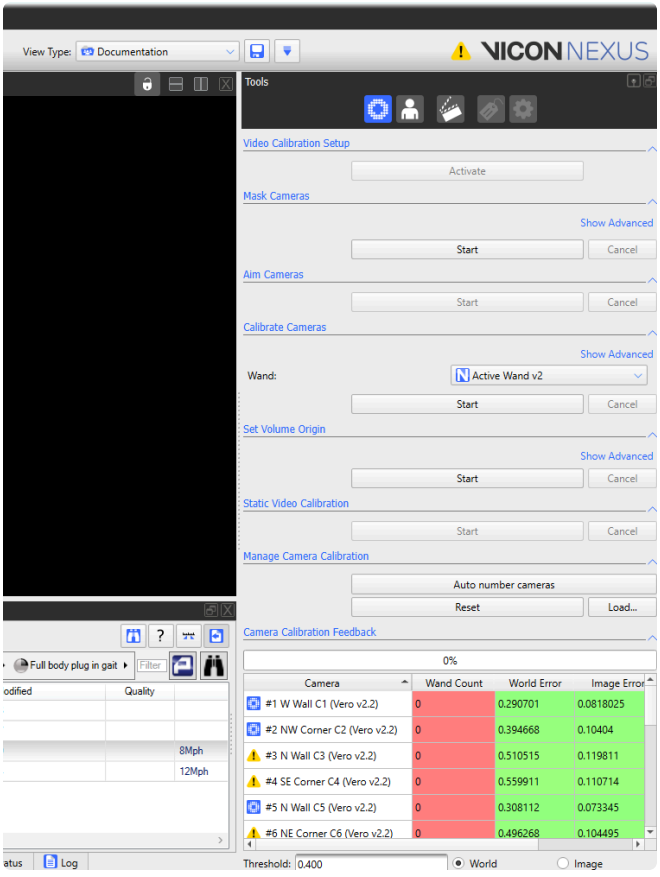
For calibration and system preparation purposes you must ensure the system is in **live mode**.



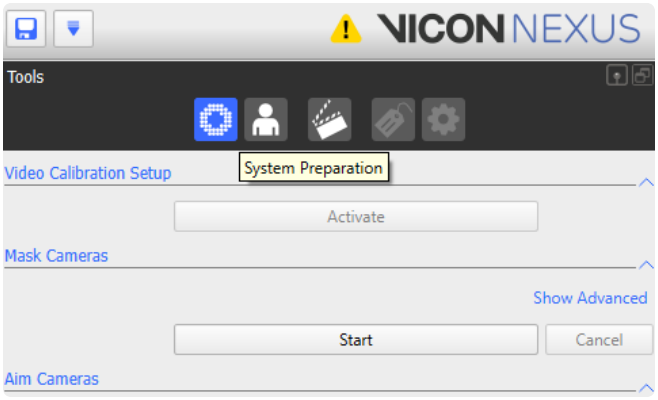
Vicon camera display state based on offline (left) and live (right) mode.

If your cameras are not currently displaying under the '**Vicon Cameras**' device group, click the '****Go live***' button to change system state.

Once you are ready to calibrate your cameras, look to **tools** panel on (by default) the right hand side, and make sure the **system preparation** tab is selected.



Tools panel, in system preparation view



How to select the system preparation tab from the tools panel

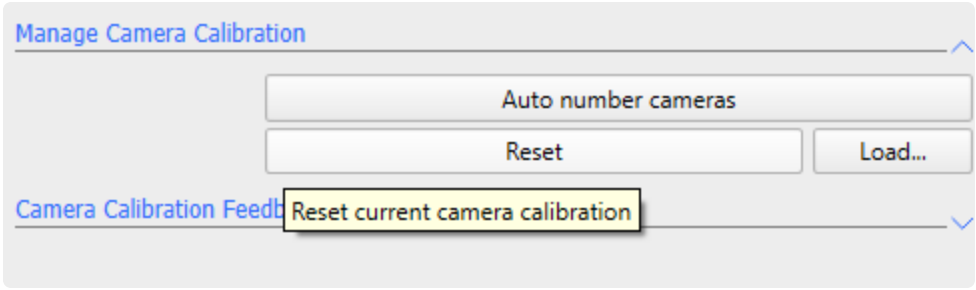
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Resetting Calibration

In order to ensure previous calibrations do not influence the day's calibration, you will need to remove the calibration profile from every camera.

If this is not a brand new calibration, do not reset camera calibration.
This will remove all previous camera information about their relative position to each other and the volume.

At the bottom of the **system preparation** tab you will see the **manage camera calibration** section.



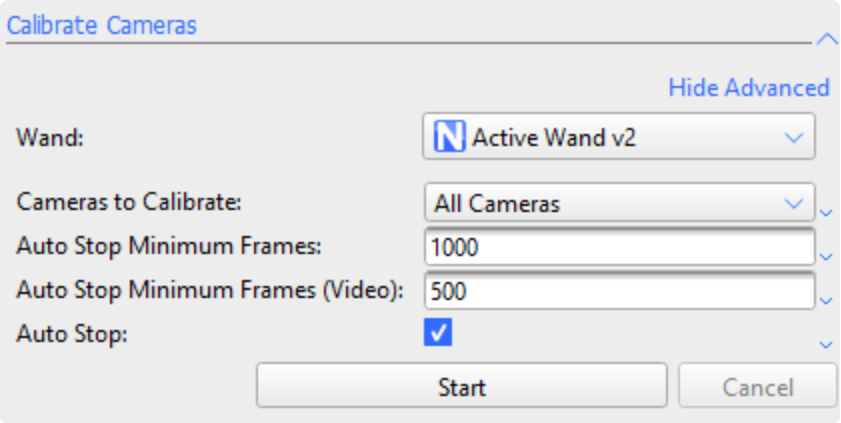
Click the **Reset** button in . If you are in 3D Perspective view, and the cameras were previously calibrated, you will see the cameras either disappear or reposition into a straight line (depending on view option selected).

This represents that the cameras no longer know where each other are, relative to themselves.

Do not select 'auto number cameras' - this will reassign camera IDs and put them out of sync with the lab's physical labelling.

Calibration Parameters

Underneath the '**Calibrate Cameras**' section you will see "*Show/Hide Advanced*" - ensure this is toggled to display all calibration properties.



Calibrate cameras section with all properties displayed

The properties here are as follows:

- **Cameras to Calibrate**: *whether to calibrate all or specific cameras.*
- **Auto Stop Minimum Frames**: *the minimum number of frames a camera has to capture to end calibration.*
- **Auto Stop Minimum Frames (Video)**: *as above, but for video cameras - not relevant to this guide.*
- **Auto Stop** (*toggle*): *whether to use the auto-stop functionality above.*

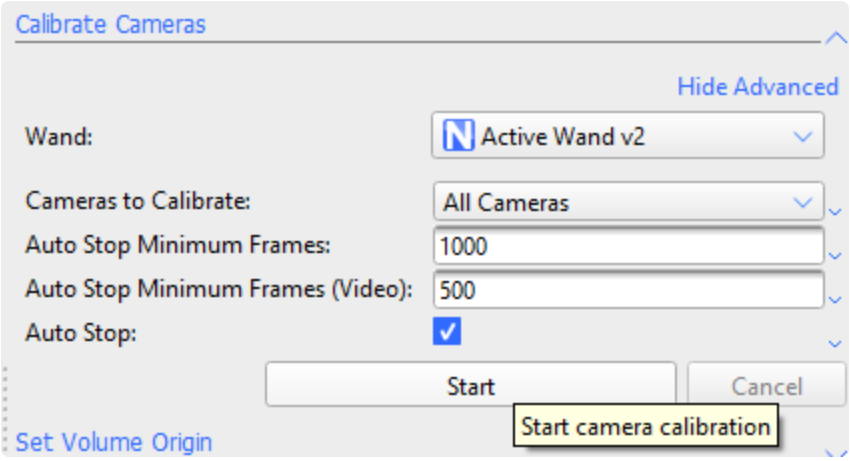
If this is your first calibration, use the following presets:

- **Cameras to Calibrate**: All Cameras
- **Auto Stop Minimum Frames**: 1000
- **Auto Stop**: Yes ☑

For more information on recalibration, calibrating individual cameras, or auto-stop:

check the advanced calibration configuration guide.

Once you are happy with these settings, start your camera calibration by clicking the *Start* button.

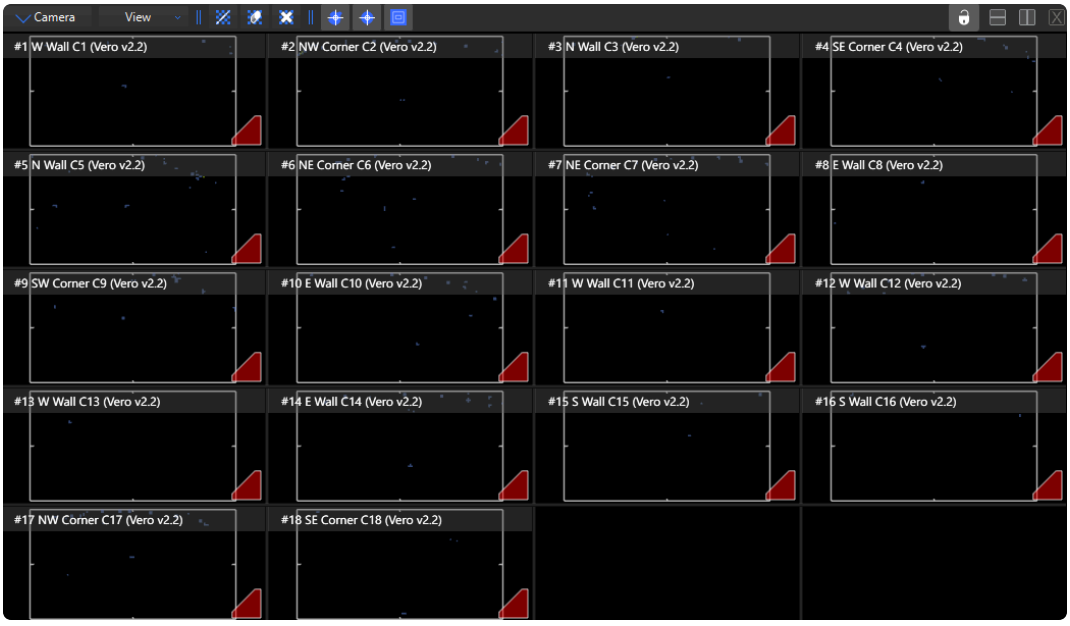


Position of the calibration start button

Calibrating with the Active Wand

Once you begin your your calibration the Nexus window will automatically switch to display the camera view for every camera registered.

This includes disabled, disconnected, and missing cameras. When calibrating with '**all cameras**' ensure the cameras present in the System/Resources view are the cameras you wish to calibrate.



View of all cameras ready to begin calibration

Once you begin calibration the camera lights within the room will begin to flash blue, indicating they are observing calibration frames.

TODO add camera state colour images/gifs

As the cameras accept calibration frames they will begin to flash quicker, until switching to a solid green hue once they have reached their minimum frame value for the auto-stop setting.

Using the defaults given in the previous part, this would be after they have tracked 1000 frames of the wand.

TODO add camera state colour images/gifs

Ensure the active wand is in '**strobe**' not '**continuous**' mode.

TODO add active wand annotated image

Wand Waving

To calibrate the cameras you must wave the wand within the volume, so that they are able to see the wand LEDs, until all cameras receive enough frames to reach their auto-stop minimum. This section covers the minimum you will need to know to calibrate your system; the dedicated wand waving guide covers this topic in more detail.

To ensure a successful calibration:

- Start from the inside of the volume, and 'paint' the space.
- Move outwards from the inside and attempt to cover
- Hold the wand away from your body, as to avoid occlusion from your body.
- Turn your body so that different cameras are able to see the wand LEDs.
- Wave the wand held flat, with the LEDs pointing upward, facing towards the camera.
- Move the wand in a circular motion, ensuring that you move the wand up and down to cover the volume.

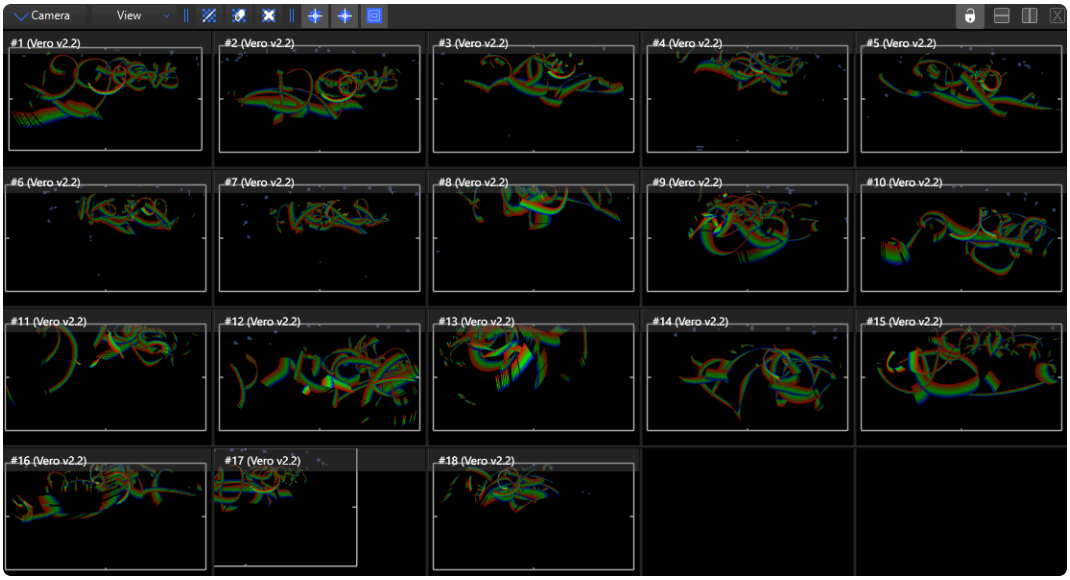
Your goal whilst calibrating is to provide as much visual data to the cameras as possible, covering the space and ensuring as many cameras as possible see the wand within every area of the volume.

Do not spend disproportionate time on one area of the volume until the volume has been fully covered. Calibration occurs on a per-camera basis, and any camera that is not effectively calibrated will provide misleading information for other cameras.

Successful Calibration

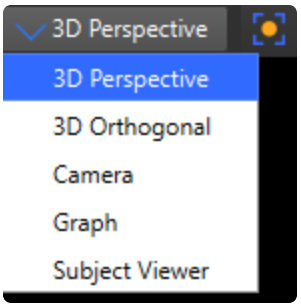
Once you have completed calibration, all camera lights will change from green to blue, and Nexus will begin the calibration feedback process.

The Nexus window will continue to display the wand wave trail data until you leave this view. This is a visual representation of where the cameras saw the wand whilst you were calibrating.

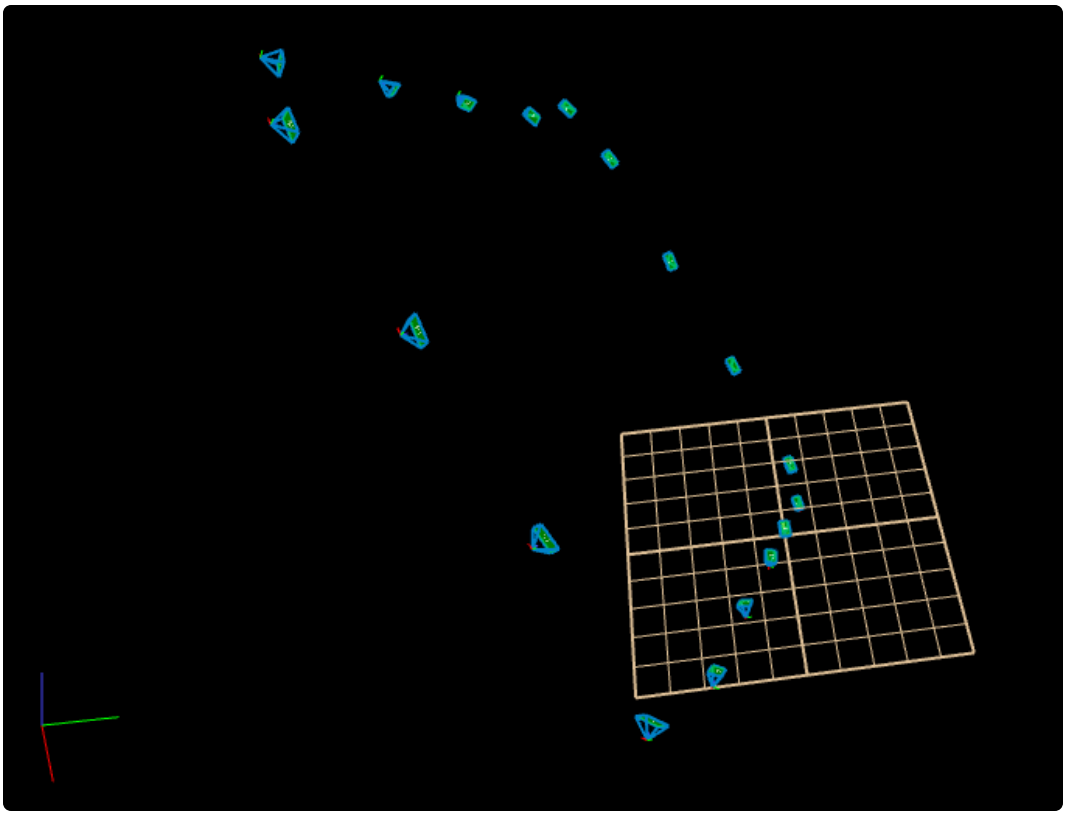


Wand wave data output

If you switch to '3d perspective' view you will be able to see immediate effects from the calibration.



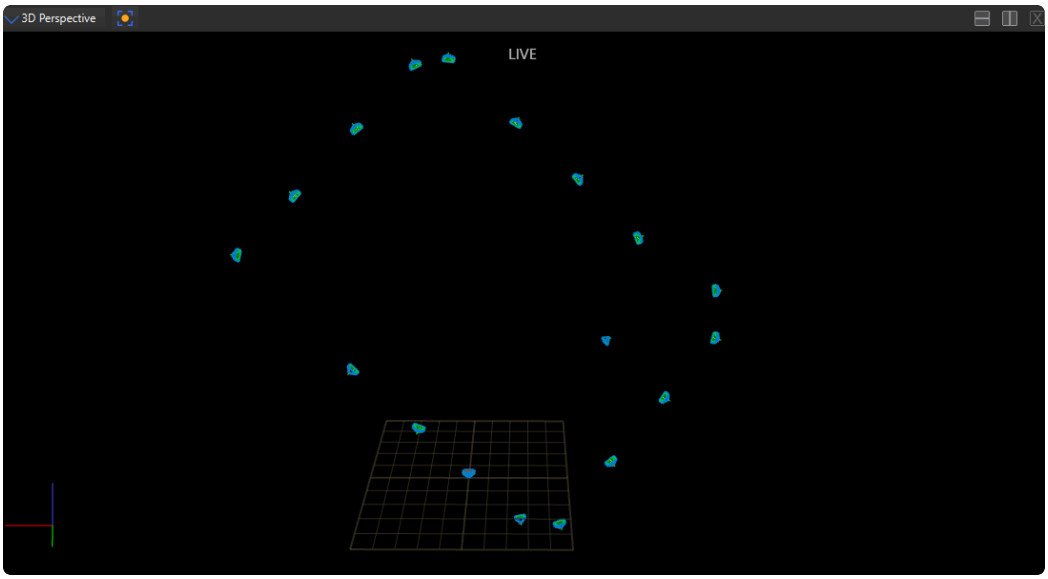
Once you are in 3D perspective view, note the position of each camera in the software, relative to each other.



View of a successful calibration - all cameras are spaced as per setup in the room.

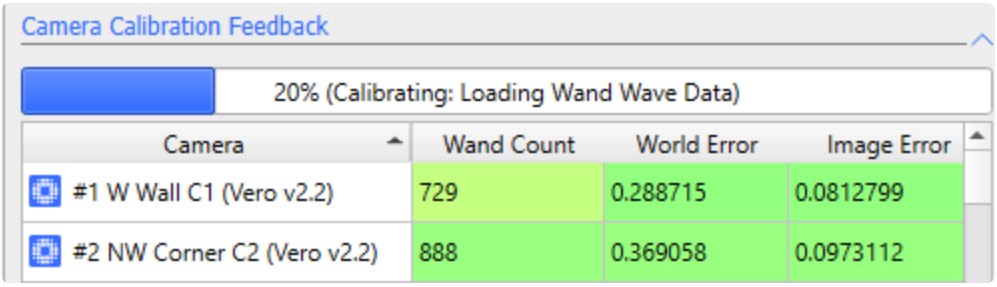
If this is representative of their position to each other, the calibration was successful for determining camera position.

This is not a guarantee of successful calibration for capture, but it is the minimum to expect at this stage. If any camera is unexpectedly out of position, consider that the calibration needs to be reattempted.

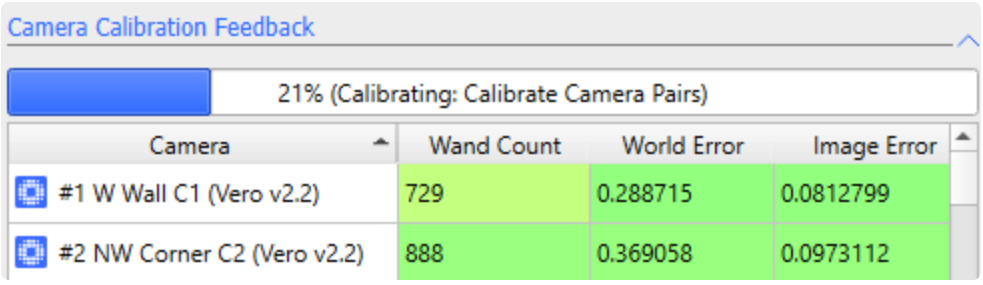


View of an unsuccessful calibration - one camera is isolated, away from where it should be within the camera ordering.

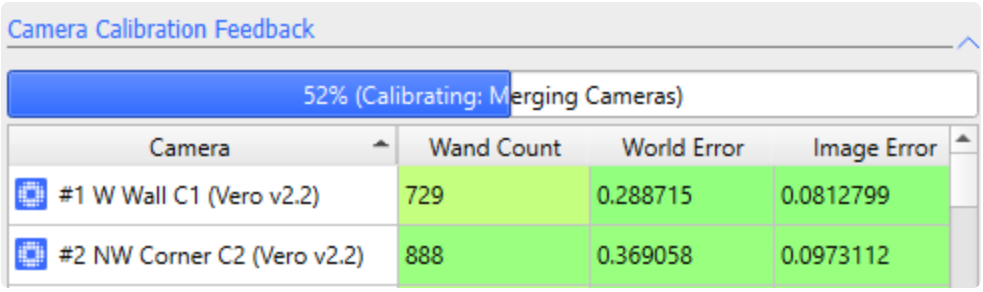
At the bottom of the **tools panel**, underneath the *camera calibration feedback* section, you will see a progress bar. This is Nexus determining the camera calibration from the wand wave data you have just provided.



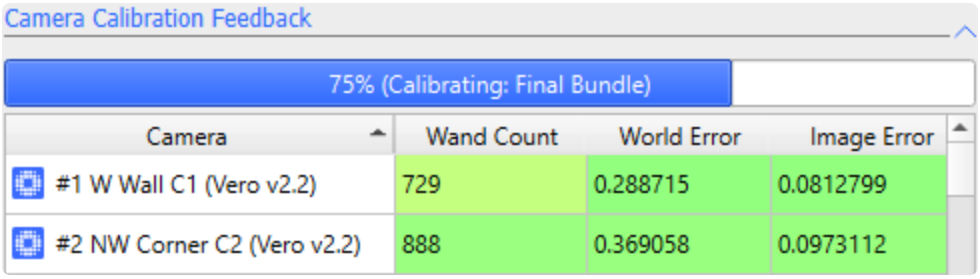
Calibration feedback initial state



Calibration feedback progressing



Calibration feedback merging information from cameras



Calibration feedback readying to output data

Once this has been completed you are ready to set the volume origin and floor plane.

Quick-Start Navigation

Previous	[Quick Start Part 1/3: Camera Masking]
Next	[Quick Start Part 3/3: Volume Origin and Plane]
